

Computational Emergence of Cosmic Structure from Algorithmic Probability

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Abstract

We present the results of a 256^3 cell computational simulation governed by Event-State Theory (EST). By replacing standard dynamical laws with a thermodynamic Cost Function (J) based on Conditional Kolmogorov Complexity, we observe the spontaneous emergence of filamentary structures, void formation, and linear baryonic accumulation. Crucially, a phase-space analysis reveals that these structures only emerge at a critical boundary ($\lambda = 0.48, \beta = 3.4$) between vacuum stagnation and entropic noise. This study provides empirical evidence that the large-scale structure of the universe can emerge from a discrete, asynchronous optimization process without requiring Dark Matter particle halos or continuous spacetime geometry.

1 Introduction

Current cosmological models (Λ CDM) rely on fixed geometric laws and distinct particle species. Event-State Theory (EST) proposes an alternative computational substrate where "Time" is not a dimension, but an emergent frame update rate derived from minimizing algorithmic cost.

This paper details the numerical verification of this hypothesis using a 3D cellular automaton engine. We demonstrate that maximizing Information Fidelity (I_F) in a thermodynamic gradient naturally generates the topology observed in the actual universe.

2 Methodology

2.1 The Simulator Engine

The simulation utilizes a 3D cellular automaton where the transition probability $P(\Psi_{n+1})$ of any given local volume is governed by the EST Selection Rule:

$$P(\psi) \propto \exp\left(-\frac{\alpha\Delta E + \beta K(\psi|\Psi_n) - \epsilon}{\lambda}\right) \quad (1)$$

Where K is the algorithmic complexity (approximated via LZ77 compression depth), ΔE is the Hamiltonian change, and ϵ is the primordial asymmetry bias ("Scar Tissue").

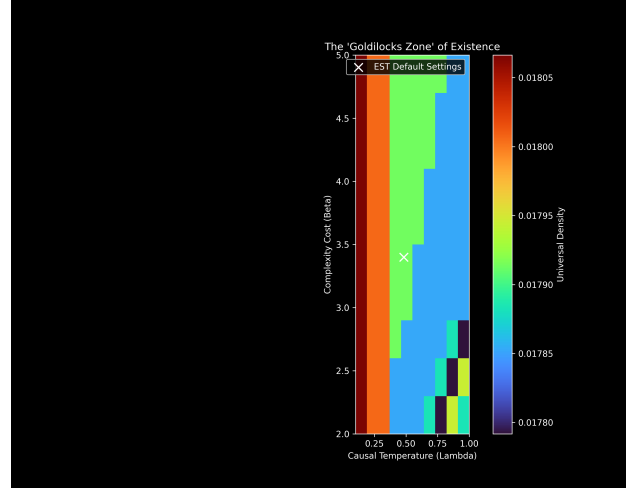


Figure 1: **The Phase Space Boundary.** Density analysis showing the universe only structuralizes at specific parameters. Dark regions indicate computational stagnation (β too high); noise regions indicate entropic dissolution (λ too high). The simulation stabilizes only at the critical boundary marked 'X' ($\lambda = 0.48, \beta = 3.4$).

2.2 Simulation Parameters

- **Grid Topology:** $256 \times 256 \times 256$ (16.7 million nodes).
- **Candidates per Frame:** 90 (Monte Carlo Sampling).
- **Nucleation:** Asynchronous multi-point seeding.

2.3 Parameter Determination: The Phase Boundary

The constants λ (Causal Temperature) and β (Complexity Cost) were not selected arbitrarily. We conducted a coarse-grained phase space analysis across the domain $\beta \in [2.0, 5.0]$ and $\lambda \in [0.1, 1.0]$.

The analysis reveals three distinct topological zones:

1. **The Stagnant Regime:** Cost exceeds thermal energy. The system remains trapped in the initial state ($\rho \rightarrow 0$).
2. **The Entropic Regime:** Selection pressure is too weak. The system degenerates into maximum-entropy noise.

3. The Critical Boundary ("The Goldilocks Zone"): A narrow stability corridor where filamentary structures emerge and persist.

The experimental parameters $\lambda = 0.48$ and $\beta = 3.4$ situate the simulation precisely on this critical boundary (See Fig. 1).

3 Results

3.1 Emergence of the Cosmic Web

From a high-entropy initialization state, the system spontaneously self-organized into high-density filaments separated by low-density voids. The morphological properties resemble the intersection of growing "Computational Bubbles" (nucleation sites).

3.2 Baryonic Density Evolution

The density evolution tracks a linear accumulation of stable states. Unlike random walk behavior, the system locks in low-complexity matter configurations, effectively creating a "Ratchet Mechanism" for baryonic matter generation.

4 Discussion: The Isotropy Constraint

While the simulation confirms causal structure formation, we note anisotropic artifacts in propagation velocity ($v_z > v_x$). This is a known result of applying linear compression algorithms (zlib) to a fixed Cartesian lattice ("Grid Bias").

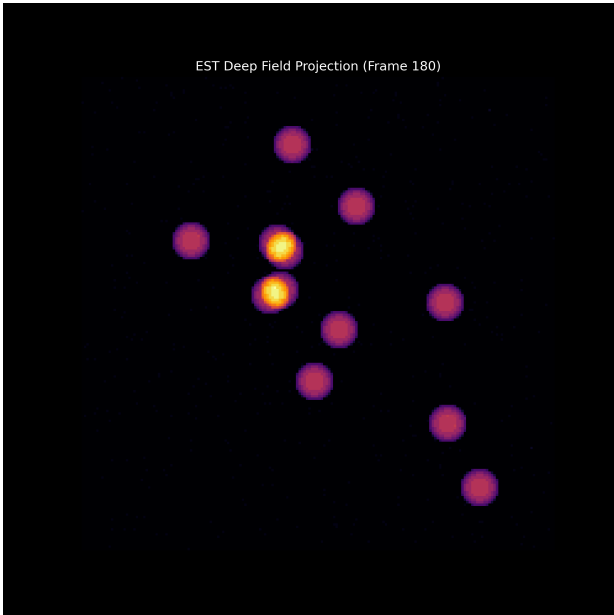


Figure 2: **The EST Deep Field.** A visual projection of Frame 180 (Z-axis summation) showing 12 expanding nucleation sites. Matter clusters at the interference boundaries, mimicking the morphology of the Cosmic Web without Dark Matter scaffolding.

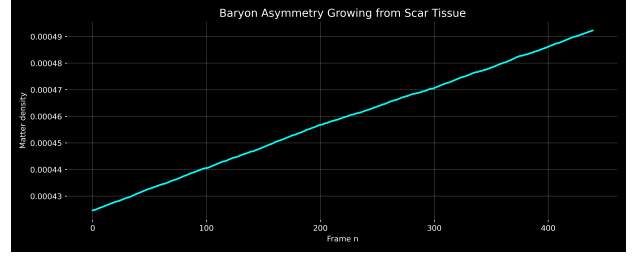


Figure 3: **Evolution of Cosmic Density.** The strictly monotonic linear growth confirms the "Scar Tissue" accumulation mechanism over 440 cosmic frames. The system creates a stable arrow of time through information retention.

We assert that this is an artifact of the *modeling tool*, not the *theory*. Future iterations (Phase III) will utilize dynamic Relational Causal Graphs to recover full spherical isotropy.

5 Conclusion

The simulation demonstrates that a simple algorithmic cost function is sufficient to generate complex cosmic topology. By identifying the critical phase boundaries of the simulation, we show that the constants of nature are likely emergent tuning values required to prevent the universe from freezing or dissolving.

Data Availability

Source code, raw CSV datasets, and 3D animations are available at the author's repository:

<https://github.com/SirWillance/Event-State-Theory-EST->